

VANET Cluster-on-Demand Minimum Spanning Tree (MST) Prim Clustering Algorithm

Jerry John Kponyo , Yujun Kuang , Enzhan Zhang , Kamenyi Domenic
Mobilelink Lab, University of Electronic Science and Technology of China
Chengdu, China

jjkponyo.soe@knust.edu.gh, kyj@uestc.edu.cn, zhangenzhan@126.com, dkamenyi@yahoo.co.uk

Abstract—Vehicle to Vehicle (V2V) communication offers great potential as far as information dissemination in VANETs is concerned. Research has shown that clustering vehicles and relaying information through cluster-heads (CHs) has several advantages over allowing all the vehicles to broadcast the information. Forming and maintaining stable clusters as well as ensuring good QoS in intra-cluster communications has always been a great challenge. In this paper we present a VANET Cluster-on-Demand (CoD) Minimum Spanning Tree (MST) Prim algorithm which clusters vehicles taking into consideration the intra-cluster QoS. Matlab simulation results of the algorithm applied to real traffic data has shown the algorithm's ability to successfully form clusters with good QoS. The algorithm has also been shown to compare very well with Dijkstra's algorithm which is one of the best clustering algorithms.

Keywords—Dijkstra's algorithm, Minimum spanning tree (MST), Prim algorithm, Vehicular ad hoc networks (VANETs)

I. INTRODUCTION

The characteristics of VANETs imposes many challenges and constraints on their implementation. To begin with, vehicles moving at variable velocity in the network, change the network topology frequently which leads to density variations at a considerable rate. Secondly the frequently changing nature of traffic results in large inter-vehicular gaps. In order to overcome such scenarios, clustering has been proposed as a potential concept for implementing many applications in VANETs. Clustering in VANETs is the process of organizing vehicles into groups based on some specific common characteristics. One among the vehicles is elected as the cluster-head (CH) and is responsible for coordinating the members of the cluster. According to [1] clustering has many benefits, such as optimizing the bandwidth utilization and allocation of resources to cluster members thus avoiding retransmission. Cluster stability is the major concern in clustering techniques. Cluster reconfiguration, CH changes and cluster maintenance are major challenges in VANETs, so the major criteria for cluster formation is to form stable clusters. In this paper we present a VANET Cluster-on-Demand (CoD) Minimum Spanning Tree (MST) Algorithm which forms clusters by taking the intra-cluster quality of service (QoS) into consideration. In many VANET cluster-based algorithms the major task is the formation and maintenance of the clusters, however; in our algorithm the clusters are formed only for the purpose of computing and relaying information on traffic density within

the coverage area of the cluster-head. The clusters are then dissolved immediately after the information has been successfully transmitted. Since there is no need to maintain the clusters and consequently the cluster-heads after the information has been relayed, the difficulty of maintaining stable clusters does not arise in this case. The Prim Algorithm which is used in implementing the MST also makes sure the size of the cluster formed is not too large to affect the QoS of information transmission in the cluster. The rest of this paper is organized as follows: Section II discusses related work. Section III gives a detailed presentation of the clustering algorithm. Simulation of the algorithm and analysis of results is tackled in section IV and conclusion and future work is dealt with in section V.

II. RELATED WORK

Within the past decade many VANET researchers have focused on developing effective ways of getting cars to communicate among themselves. Clustering for information dissemination has become a very effective technique and as such a lot of emphasis has been placed on developing effective and efficient algorithms for clustering. Clustering techniques like lowest Id [2] and highest degree [3] algorithms have been proposed in the past but in both cases the cluster-head (CH) election leads to more clustering and has been found to result in unstable clusters. Other clustering algorithms like [2], [4] which are beacon based form clusters using mobility metrics and the received signal strength (RSS). The effect of multipath fading losses in the signal strength calculation is ignored and this has an adverse effect on the stability of the clusters formed by this algorithm. Authors in [5] proposed a position based clustering algorithm in which the road is divided into several cells with each cell having an anchor point. The vehicles in the cells closest to the anchor points are chosen as the CHs until they leave the cell. Clusters formed by this method have been found to be unstable. A modified version of Distributed and Mobility Adaptive Clustering (DMAC) algorithm [3] is proposed for VANETs in [6]. The modified algorithm deals effectively with the problem of rapid changes in network topology as a result of the mobility of vehicles. A weight is assigned to vehicles based on position, connectivity, mobility etc. and the vehicle with the highest weight is chosen as the CH. Although the clusters formed by this algorithm are found to be stable there is an increase in overhead for the network. In [7] the road is broken into cells and vehicles

travelling in the same direction are grouped into clusters. The first vehicle to enter a particular cell is chosen as the CH. With the vehicles travelling at variable speeds, frequent cluster-head changes creates a lot of overhead for the network. [8] gives a modification of [7] where clustering is based on the distance and direction the vehicle takes after crossing the junction. This algorithm shows better stability but the clustering is restricted to the region around the intersection. Information dissemination is therefore not possible with this algorithm. [9] proposes a cluster-based information dissemination algorithm and also calculates the density of vehicles approaching an intersection beforehand and sends the information to a traffic signal so it can automate traffic control which helps in considerably reducing the waiting time at a traffic intersection. [13] proposes the dijkstra's algorithm for cluster formation and anchor point selection in vehicular ad hoc networks. It is evident that in all the algorithms discussed above the clusters are active all through the entire duration of travel and as such network resources are needed to maintain the clusters. It can also be seen that not much attention is paid to the QoS of intra-cluster communication. In this paper we propose a cluster-based algorithm which uses Prim Algorithm to implement MST in forming clusters. This algorithm takes into consideration the QoS of intra-cluster communication by calculating a maximum cluster size for every cluster beyond which the quality of communication within the cluster is not acceptable. The clusters are also formed only for the purpose of information dissemination and are dissolved immediately after the information is relayed and is only reactivated at the next snapshot of the network.

III. CLUSTERING ALGORITHM

A. Junction Scenario

Figure 1 shows a junction scenario for the cluster-based VANET clustering algorithm. As can be seen from Figure 1, no clusters are permitted to form within a radius of $2 \times d_{th}$ from the junction node. The vehicles can form and join clusters only when they are d_{th} away from the junction node. It is also important to mention that the clusters are formed only when information has to be relayed to the junction nodes. Reference tags are placed on the road to prompt vehicles leaving a junction to begin to join clusters and those entering a junction not to join any cluster.

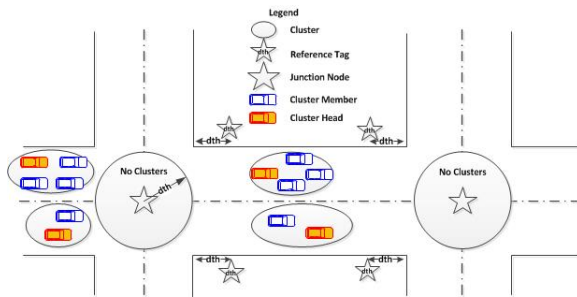


Fig. 1. Junction Scenario for Cluster-Based VANET

B. Cluster-on-Demand MST Clustering Algorithm

The position of vehicles is obtained using GPS. A vehicle in a specific road segment is selected at random and using Prim minimum Spanning Tree algorithm, a list of vehicles are selected to form a cluster. The selected vehicle becomes the cluster-head for that specific cluster. In selecting the cluster members the criteria is to select vehicles with similar velocities, near the cluster-head and moving in the same direction as the cluster-head. Also the number of vehicles is regulated so that the maximum area and the number of vehicles per cluster are not exceeded. Our stopping criteria is the area covered by the cluster since as we increase number of vehicles under one cluster, the area increases until it reaches a threshold where QoS degradation is unacceptable. Prim algorithm [10], [11] is used to find a minimum spanning tree for a connected weighted graph. The algorithm finds a minimum cost spanning tree by selecting edges from the graph one-by-one as follows:

- 1) It starts with a tree, T consisting of a single starting vertex, x .
- 2) Then, it finds the shortest edge emanating from x that connects T to the rest of the graph (i.e., a vertex not in the tree, T).
- 3) It adds this edge and the new vertex to the tree T .
- 4) It then picks the shortest edge emanating from the revised tree T that also connects until a minimum spanning tree satisfying a specified condition is achieved.

For example, consider a graph G formed by vehicles in motion in a road segment where vertices v is the total number of vehicles in this road segment and moving in the same direction. Edges

$$e = v - 1 \quad (1)$$

is the number of edges in the graph G without forming a loop, and edge weights w is the metric resulting from combining similarity of velocity and minimum distance. Let the accepted QoS be represented by k . k is calculated from the distance D of any nearest vehicle from the cluster-head and the maximum range of the cluster-head R . We want to generate a graph T to form a cluster that satisfies our desired QoS. Starting with an empty graph T , we randomly select a vehicle in a road segment. Let this vehicle be v_1 . Starting from v_1 , we let vertex v_1 and edge e_1 have the smallest metric weight (w) in graph G , we remove (e_1, v_1, w_1) from G and add e_1 and v_1 to T . For the remaining edges (v_i, e_i) , we check if v_i is not already in G and if so, we find the minimum weight w_i in graph G then add v_i and e_i to T . We then repeat this until QoS condition of k is achieved. We then check if the number of vehicles so far selected exceeds the allowed value n . If so, we reduce the number of vehicles to satisfy this condition starting with the last vehicle to be added. The cluster is then formed with v_1 as the cluster-head. Details of the algorithm are as depicted in Algorithm 1.

IV. SIMULATION AND ANALYSIS OF RESULTS

A. Evaluation Metrics

We test our Algorithm using the following metrics:

Algorithm 1 VANET MST Graph (T) Algorithm

Require: <Vehicles in a road segment forming Graph G>

Ensure: <Cluster Graph (T)>

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1: let  $G$  be the graph containing candidate vehicles that can be clustered together with
   the selected vehicle in a road segment;
2: let  $T$  be an empty graph;
3: let  $R$  be the maximum range of the cluster-head;
4: let  $R'$  be the distance between the candidate vehicle and the cluster-head;
5: while ( $R' < R$ ) do
6:   let  $(v, e)$  have the smallest metric weight ( $w$ ) in graph  $G$ ;
7:   remove  $(v, e, w)$  from  $G$ ;
8:   add  $v$  and  $e$  to  $T$ ;
9:   for each edge  $f = (v_i, e_i)$  with weight  $w_i$  do
10:    if  $v_i$  is not already in  $T$  and is not a candidate for selection by another
       cluster-head then
11:      find edge  $g = (u_i, g_i)$  with weight  $z_i$  in  $G$ ;
12:      if edge weight ( $f$ ) < edge weight ( $g$ ) then
13:        replace edge weight  $g$  with edge weight ( $f$ );
14:      end if
15:    end if
16:  end for
17: end while
18: return Cluster

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- 1) The distance between cluster-head vehicle and any other nearest vehicle is given by:

$$D = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2} \quad (2)$$

where x and y are the Latitude and Longitude respectively of the two vehicles.

- 2) Quality of service(QoS) depends on the maximum allowable distance between the cluster-head vehicle and cluster members. The greater the distance the less the QoS. Let the maximum allowable distance for attaining acceptable QoS be R . This represents the Intra-cluster transmission range of a cluster-head. Let D be the distance between a cluster-head vehicle and a member of the cluster calculated using equation(2). Thus, QoS is given by:

$$QoS = \frac{R}{D} \quad (3)$$

In this regard we consider the following:

- Total length of the road segment = y_{meters}
- Total number of vehicles expected on the road = N
- Intra-cluster transmission range of cluster head = R
- Inter-cluster transmission range of cluster-heads = $2R$
- The threshold distance $d_{th} = 2R$

$$\text{Number of clusters}(q) = \frac{(y - 2d_{th})}{2R} \quad (4)$$

$$\text{Number of vehicles per cluster}(n) \leq \frac{N}{q} \quad (5)$$

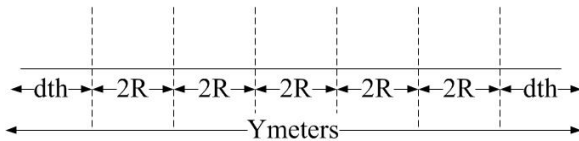


Fig. 2. Road Segmentation

TABLE I
TRAFFIC IN ONE KILOMETER ROAD SEGMENT

Snapshot	New York City	Disney World	State Fair
1	455	345	200
2	470	347	195
3	499	344	190
4	480	337	192
5	470	340	199
6	488	339	194
7	495	345	190
8	498	347	192
9	489	339	196
10	497	340	199

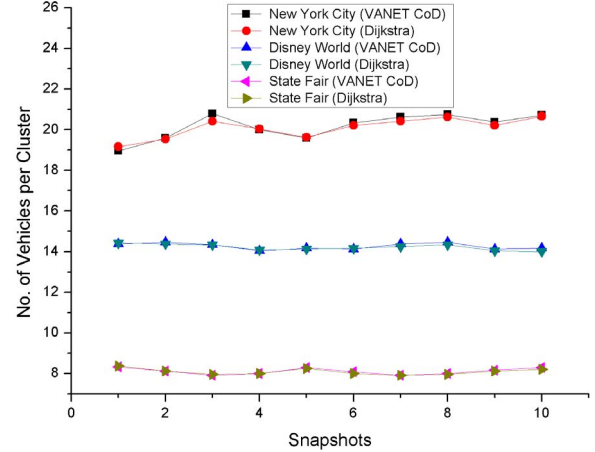


Fig. 3. Number of vehicles in a cluster per snapshot

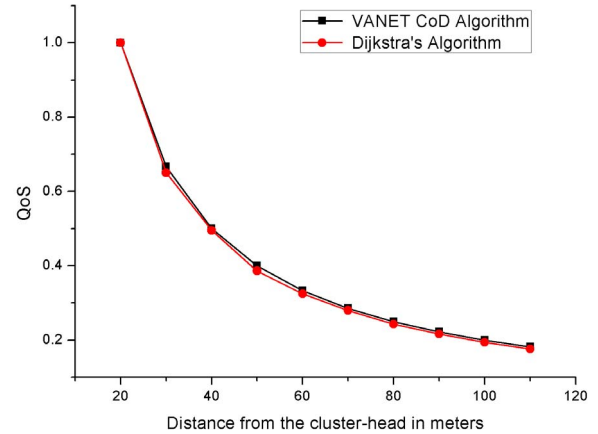


Fig. 4. QoS Evaluation

B. Experimental Setup

We adopt the real world vehicular mobility trace data collected from three different locations from CRAWDA [12] to generate our road segment traffic. We record traffic in a one kilometer road section in New York City (with high traffic), Disney World, Florida (with medium traffic) and State Fair (less traffic). We select vehicles travelling with similar velocities, and for each vehicle selected, we record the value of X (latitude) and Y (longitude) co-ordinates. We then select one

vehicle at random to act as a cluster-head. Using the recorded X and Y co-ordinates, we calculate the distance between a vehicle and the cluster-head using formula (2). To achieve optimized selection, we deploy Prim algorithm to come up with a MST of vehicles closest to the vehicle randomly chosen as a cluster-head. We take snapshots every 30 seconds setting the maximum allowable distance between the cluster-head vehicle and cluster members at 20 meters. We observe the first 10 snapshots. With simulated data, we implement our algorithm on Matlab using a laptop with 2 GB memory and a Core (TM) 2 Duo 2.00GHz Intel processor.

C. Evaluation of Results

Table I shows traffic in one kilometer stretch of a road segment for the three areas. New York City has more traffic than Disney world and State fair. Using equation(4), we calculate the number of clusters in a 1000 meter road segment with $R=20$ meters (24 clusters). We utilize equation(5) to calculate the maximum number of vehicles per cluster and use prim algorithm to select vehicles nearer to the cluster-head to form the cluster. The number of vehicles per cluster in each scenario is plotted in figure 3. It is observed that in New York City, more vehicles are selected per cluster due to high traffic. Similarly fewer vehicles are selected per cluster in State Fair. Figure 3 also compares the results from our proposed clustering scheme to clustering using the dijkstra's algorithm and it is observed that the results compare very well with those of the dijkstra's algorithm. The advantage of our algorithm over the dijkstra's algorithm is that whereas our algorithm can be used for cases where the weights are negative i.e. for cases where the vehicles move in opposite directions, the dijkstra's algorithm can only be used for cases where the weights are positive. As can be seen from figure 4 it is clear that as vehicles move beyond the cluster-head range, the QoS become poorer. It can be seen from figure 4 that the QoS degrades to some unacceptable level as the vehicles move beyond 50meters of the cluster-head. For maximum performance therefore the algorithm always limits the clusters to vehicles within the maximum range of the cluster-head.

V. CONCLUSION AND FUTURE WORK

In this paper we have introduced the idea of forming clusters on demand as a means of checking the maintenance and stability of clusters which has been a major issue as far as V2V communications in VANETs is concerned. Matlab simulation results of the algorithm have shown the clustering algorithm to work successfully as far as cluster formation is concerned. The results have also shown that beyond a distance of 50meters from the cluster-head the Quality of Service of intra-cluster communications is poorly affected. This means that for maximum performance the algorithm always limits the vehicle distance to the maximum range of the cluster-head. Future work will focus on using the concept of cluster-on-demand to develop an effective traffic information dissemination scheme through which drivers could query junction nodes to obtain the status of traffic at various segments of the road.

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